

Relation, Attribute, Tuple and Domain in DBMS

with Characteristics of Relations

1. Introduction

The **Relational Model**, proposed by E. F. Codd, is the most widely used data model in database management systems. In this model, data is represented in the form of **relations (tables)**. Each relation consists of **attributes**, **tuples**, and **domains**, which together define the structure and meaning of stored data.

Understanding these basic terms is essential for studying relational databases, SQL, normalization, and relational algebra.

2. Relation

2.1 Definition

A **relation** is a collection of related data items organized in the form of a **table**, consisting of rows and columns.

Formally, a relation is defined as a **subset of the Cartesian product of domains**.

In simple terms:

- A relation represents a **table**
 - Rows represent records
 - Columns represent fields
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2.2 Example of a Relation

STUDENT Relation

RollNo	Name	Department	Marks
1	Amit	CS	85
2	Ravi	IT	65
3	Neha	CS	92

Here:

- Relation name $\rightarrow$ STUDENT
 - Rows $\rightarrow$ Tuples
 - Columns $\rightarrow$ Attributes
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2.3 Explanation

Each relation has:

- A **unique name**
- A fixed set of attributes
- A set of tuples representing real-world entities

Relations are the fundamental building blocks of relational databases.

3. Attribute

3.1 Definition

An **attribute** is a **named column** in a relation that represents a property or characteristic of an entity.

Each attribute describes a specific piece of information about the entity represented by the relation.

3.2 Example

In the **STUDENT** relation:

Attribute	Meaning
RollNo	Student identification number
Name	Student name
Department	Department of student
Marks	Marks obtained

3.3 Explanation

- Attributes define the **structure (schema)** of a relation.
 - Each attribute has a **name** and a **domain**.
 - Attribute values in a tuple must belong to the domain of that attribute.
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4. Tuple

4.1 Definition

A **tuple** is a **single row** in a relation that represents one complete record or instance of an entity.

Each tuple contains one value for every attribute of the relation.

4.2 Example

From the **STUDENT** relation:

RollNo	Name	Department	Marks
1	Amit	CS	85

This row is **one tuple**.

4.3 Explanation

- A tuple represents a **real-world entity** (e.g., one student).
 - Tuples are unordered.
 - No two tuples in a relation can be identical.
 - The number of tuples in a relation is called its **cardinality**.
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5. Domain

5.1 Definition

A **domain** is a **set of atomic values** from which attribute values are drawn.

It specifies the **type, format, and range** of values that an attribute can take.

5.2 Example

Attribute	Domain
RollNo	Integer
Name	Character string
Department	{CS, IT, EE, ME}
Marks	Integer (0–100)

5.3 Explanation

- Domains ensure **data integrity**.
 - They prevent invalid data entry.
 - Each attribute must be associated with exactly one domain.
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6. Characteristics of Relations

The relational model imposes several rules or characteristics on relations to ensure consistency and correctness of data.

6.1 Unique Relation Name

- Each relation must have a **unique name** within a database.
 - This avoids ambiguity during data access.
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6.2 Distinct Tuples (No Duplicate Rows)

- A relation cannot contain duplicate tuples.
- Each tuple must be unique.

This ensures that each record represents a unique entity.

6.3 Atomic Attribute Values

- Each attribute value must be **atomic (indivisible)**.
- Multivalued or composite attributes are not allowed.

Example:

Phone numbers should not be stored as multiple values in a single attribute.

6.4 Attribute Values from Same Domain

- All values of an attribute must be taken from the **same domain**.
- Mixing data types is not allowed.

Example:

Marks column cannot contain both numbers and text.

6.5 Attribute Names are Unique

- No two attributes in the same relation can have the same name.
 - Attribute names uniquely identify columns.
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6.6 Order of Attributes is Irrelevant

- The order in which attributes appear in a relation does **not matter**.
 - Logical meaning remains the same even if column order changes.
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6.7 Order of Tuples is Irrelevant

- Tuples are stored as a set.
 - Changing the order of rows does not change the relation.
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6.8 Null Values are Allowed

- Attributes may contain **NULL values**.
- NULL represents:
 - Unknown value
 - Not applicable
 - Missing information

However, excessive use of NULLs is discouraged.

6.9 Fixed Degree

- The number of attributes (degree) of a relation is fixed.
 - Tuples must contain values for all attributes (or NULL).
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7. Summary Table

Term	Meaning
Relation	Table representing data
Attribute	Column describing a property
Tuple	Row representing a record
Domain	Set of allowed values

8. Conclusion

Relation, attribute, tuple, and domain are the fundamental concepts of the relational model. A relation organizes data in a structured form, attributes define the structure, tuples represent individual records, and domains ensure data validity. The characteristics of relations enforce data integrity, consistency, and independence, making the relational model reliable and widely adopted in database systems.